

Protein, Total

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate nutritional status; investigate edema.

In the entities which follow, the diseases listed are sometimes increased or decreased as indicated, but are not always so.

Causes of high total protein: dehydration; some cases of chronic liver disease, including chronic active hepatitis and cirrhosis; neoplasms, especially myeloma; macroglobulinemia of Waldenström; tropical diseases (eg, kala-azar, leprosy, and others); granulomatous diseases, such as sarcoidosis; diseases in which total protein is sometimes high include collagen disease (eg, systemic lupus erythematosus (SLE), and other instances of chronic infection/inflammation).

Causes of low total protein: pregnancy; intravenous fluids; cirrhosis or other liver disease, including chronic alcoholism; prolonged immobilization; heart failure; nephrotic syndromes; glomerulonephritis; neoplasia; protein losing enteropathies; Crohn's disease and chronic ulcerative colitis; starvation, malabsorption or malnutrition; hyperthyroidism; burns; severe skin disease; and other chronic diseases.

Very low total protein (<4 g/dL) and low albumin cause edema (eg, the nephrotic syndromes).

Limitations

Clinical interpretation is greatly enhanced by examination of the fractions composing total protein, when such separation is clinically indicated (ie, serum protein electrophoresis, quantitative immunodiffusion or other methods for IgG, IgA, IgM, immunofixation, immunoelectrophoresis).

Custom Additional Information

Total protein and albumin normally decrease by 5% to 10% upon recumbency, as in hospitalization. Drug effects are summarized.¹

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within four hours of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Hemolysis; improper labeling

Footnotes

1. Herbeth B, Diemert MC, Galli A. Total proteins. In: Siest G, Galteau MM, eds. *Drug Effects on Laboratory Test Results Analytical Interferences and Pharmacological Effects*. Littleton, Mass: PSG Publishing Co;1988:375-390.